

Printed cylindrical lens pair for application to the seam concealment in tiled displays

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Abstract: Seamless tiling of displays is one of the key enabling technologies for the next-generation large-area electronics applications. In this paper, we propose a facile method to demonstrate a seamless display using cylindrical lens pair (CLP) fabricated by dispenser printing method. Optical properties of the printed CLP and corresponding capability of concealing seam in the display are analyzed by a set of luminance simulation and measurement in terms of geometric parameters of the lens. The seamless display with an optimized CLP features a viewing angle of the seam concealment of 40°.

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References and links

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1. Introduction

Large-area displays ($> 100''$ diagonal) have been widely used for public information systems such as control room, transportation, retail, entertainment, and education. However, high cost and low production yield of such a large-area display are technical challenges [1, 2]. Therefore, as an alternative technology, a tiling method, which is to arrange several panels in the same plane, has been widely used due to its relatively simple implementation [3,4]. However, seam, which is the mechanical frame around the display panel, makes discontinuous images in the tiled display, as shown in Fig. 1(a). In the seam region, the display driving circuit is arranged, and the width of the seam has been narrowed to less than 1 mm due to the progress in the panel design [5, 6]. But it is impossible to completely remove the seam because it should support the panel mechanically. To solve this problem, several approaches have been proposed such as overlapping the edge of panels [7] and utilizing optical waveguide [8]. However, these methods require complicated mechanical and optical designs. Another approach is to add lens to cover the dark lines [9] between light emitting areas. This approach has also been applied to increase the power-conversion efficiencies of photovoltaic device by cloaking the contact fingers [10, 11]. By attaching lens pair onto the seam directly, light emitted from the vicinity of the seam is refracted toward the seam, thereby optically concealing the seam and showing a continuous picture, as shown in Fig. 1(b). Fabrication of proper shape of lens is typically based on conventional microelectronic technology such as thermal reflow of photoresist [12], gray-tone photolithography [13], chemically wet etching [14], direct laser writing [15], and hot embossing [16]. These methods are time-consuming and involve high process cost associated with the vacuum system, photolithography and etching [17]. Recently, direct lens fabrication using liquid materials has been employed as a promising, cost-effective method with high degree of design freedom, including inkjet [18, 19] and dispenser printing methods [20, 21]. In case of inkjet printing, there is a limitation on the use of highly viscous liquid materials ($>20 \text{ mPa} \cdot \text{s}$) [22] and it is difficult to fabricate large-area, high aspect ratio lens shape and cylindrical lens array [23], which is widely used for three-dimensional (3-D) display [24–26] and laser beam focusing [27]. Dispenser printing is an alternative method to overcome the disadvantages of inkjet printing, and suitable for obtaining low cost and high performance lens.

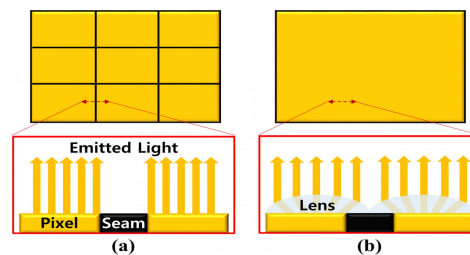


Fig. 1. Schematic diagram of (a) conventional tiled display, (b) seamless tiled display with lens pair. The insets show cross-section view of each displays.

In this paper, we propose a method for designing and fabricating cylindrical lens pair (CLP) based on dispenser printing technique for seamless display application. We analyze the luminance distribution of seamless display through simulation and measurement, followed by demonstration of seam concealment to verify that the proposed method on an actual display panel works excellently.

2. Fabrication process of cylindrical lens pair (CLP)

The method presented here provides the capabilities of engineering geometric parameters of cylindrical lens and achieving a maximum fill factor (FF) of CLP, which is important to achieve complete seam removal. In this study, we defined FF as the ratio of seam area

covered by lens to total seam area, which is determined by packing density and arrangement of each lens. If FF is not 100%, the part of seam that is not covered by the lens is observed. But all seam area is concealed by CLP due to its FF around 100% [28]. Generally, high FF enhances the optical performance of lens array in optical imaging and detection system [15]. That is why CLP was used for seamless display applications.

Figure 2 shows the fabrication process of a CLP. As a lens material, Norland Optical Adhesive 63 (NOA 63, Norland Products), which has been widely used as ultraviolet (UV) curable liquid polymer, was selected due to its high refractive index (1.56), high transparency ($\sim 99\%$ at visible wavelength), fast UV-curing property in ambient condition and low cost [29]. Especially, it has a high line pattern fidelity when continuously dispensed onto the glass substrate, showing the advantage of cylindrical lens formation. Another candidate material for fabricating cylindrical lens is polydimethylsiloxane (PDMS), which has been used for fabricating cylindrical lens array [30]. However, since PDMS has a property of rapidly spreading onto the glass at room temperature and has to be thermally cured for a long time [31], it is difficult to obtain cylindrical lens of the desired shape directly.

At first, pre-cleaned glass substrate (Soda lime, L.C.D Tec) was placed on a stage of an automatic dispenser (SHOTmini 200Sx, Musashi Engineering). The dispenser was on laboratory table with small vibration to keep it flat, thereby fixing the nozzle and the substrate. The height of nozzle was controlled by teaching pendant that can program jetting position, and it was maintained at $200\ \mu\text{m}$ in the whole fabrication process. After placing the substrate, NOA 63 was jetted continuously through metal nozzle with $900\ \mu\text{m}$ diameter in contact with the substrate in a straight path mode. With straight path mode, the length of cylindrical lens can be controlled freely within the size of the stage. In the experiment, the cylindrical lens was made $15\ \text{cm}$ long. The printed cylindrical lens was cured immediately for $1\ \text{min}$ with the $365\ \text{nm}$ wavelength UV light having intensity of $7\ \text{mW}/\text{cm}^2$ by using a portable UV lamp (LF206LS, UVITEC Cambridge). After $1\ \text{min}$ UV-curing, the stage was moved by the diameter of cylindrical lens, and subsequently formed another cylindrical lens under the same jetting condition as in the previous fabrication step, followed by full curing for $10\ \text{min}$. It is noted that both cylindrical lenses were not merged because the shape of the cylindrical lens in the first step was already fixed in a solid phase through the partial curing process. In this way, we obtained a NOA 63 CLP.

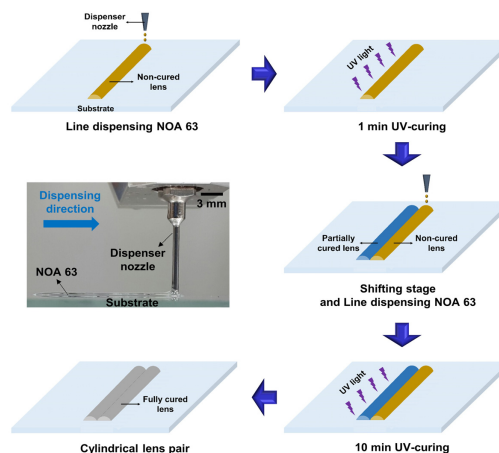


Fig. 2. Fabrication process of a CLP with the dispenser printing method. The inset shows the continuous dispensing of NOA 63.

In the fabrication process, the geometric shape of cylindrical lens could be controlled by adjusting jetting conditions. The shapes of cylindrical lens were investigated by a 3-D surface profiler (μSurf , NanoFocus) with confocal microscope mode. Figures 3(a)-3(c) show the

cross-sectional profiles of the cylindrical lens with different jetting conditions. Since both ends of the cylindrical lens are the starting and ending position of jetting where cross-section profile inhomogeneity appears, the cross-section profile at center position of cylindrical lens was measured. It was observed that each cylindrical lens showed an aspherical cross-section. The diameter of cylindrical lens increased as the stage speed decreased or jetting pressure increased. When NOA 63 was stacked by dispensing it again just above the partially cured cylindrical lens, the contact angle of cylindrical lens increased while maintaining the diameter of cylindrical lens. This means that the numerical aperture can be also adjusted through the stacking lens materials. As an example, Fig. 3(d) and 3(e) represent optical images of the fabricated CLP with a diameter of 3.058 mm, which is achieved by the moving speed of stage for 20 mm/s under the jetting pressure of 300 kPa without stacked layer. The cross-section view was captured by digital microscope (AM-413TL, Dino-Lite) after cutting the center of cylindrical lens perpendicular to the substrate using a blade. This result shows that each cylindrical lens is almost the same size and well-arranged without being merged to each other.

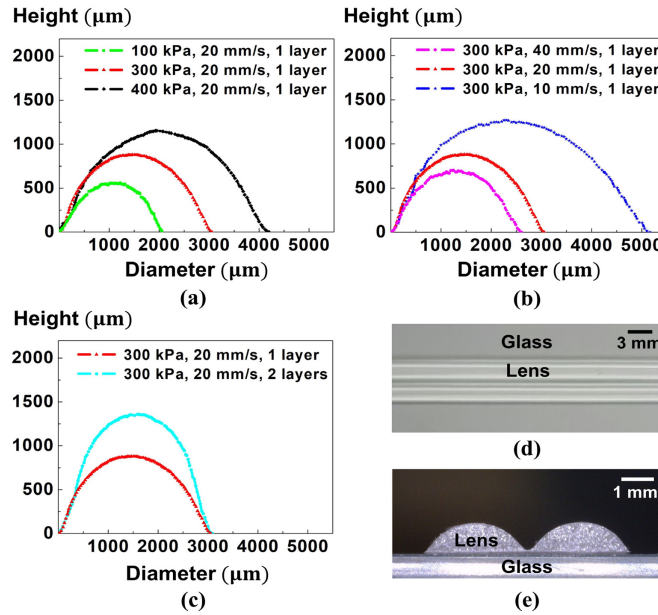


Fig. 3. Cross-sectional profiles of the cylindrical lens with different (a) jetting pressure, (b) moving speed of stage, and (c) number of stacked layer. Optical image of CLP from (d) top and (e) cross-section view.

3. Design and optimization of seamless display system

Figure 4(a) represents the image-forming behavior of a plano-convex lens. The fabricated cylindrical lens is a plano-convex lens, which generates virtual image of an object on opposite sides of the focal point when the object distance is shorter than the focal length. The image below the cylindrical lens is enlarged to cover the seam, thereby leading to the optical concealment effect. Based on Gaussian lens formula, magnification can be expressed by

$$M = \frac{f}{t - f} \quad (t < f), \quad (1)$$

where M , t and f represent the magnification of lens, the thickness of glass and the focal length of lens, respectively. It is assumed that the fabricated cylindrical lens shows the negligible difference in focal point between paraxial and marginal rays because the surface of

cylindrical lens is aspherical form [32, 33]. Also the lens formula can be applied for the aspherical refracting surface [34]. So although Eq. (1) is derived under the assumption of paraxial rays, it is suitable for the theoretical description of the seamless display system. In order to obtain the magnification, the focal length should first be calculated from the surface profile of lens by ray tracing simulation [30] or be measured by collimator [19]. Then, we can calculate the magnification of lens by inserting the focal length and the thickness of glass into Eq. (1). The viewing angle of seamless display system is defined as the theoretical maximum angular range at which seam is not seen. Figure 4(b) shows the principle of seamlessness appearance in our system. As shown in Fig. 4(b), the viewing angle is derived as the following equation:

$$\theta = 2 \arctan \left[\frac{|M|(D - 2s) - D}{2|M|t} \right], \quad (2)$$

where θ is the viewing angle, D is the diameter of lens and s is the half of seam width. The viewing angle is also related to the power of the lens which can be represented by the f -number. Generally, the f -number is defined as

$$F = \frac{f}{D}, \quad (3)$$

where F is the f -number of the lens. As a result, Eq. (2) can be rewritten in terms of F as the following equation:

$$\theta = 2 \arctan \left[\frac{t - 2Fs}{2Ft} \right]. \quad (4)$$

Thus, we can determine the viewing angle from the variables associated with a seamless display according to Eqs. (1)-(4).

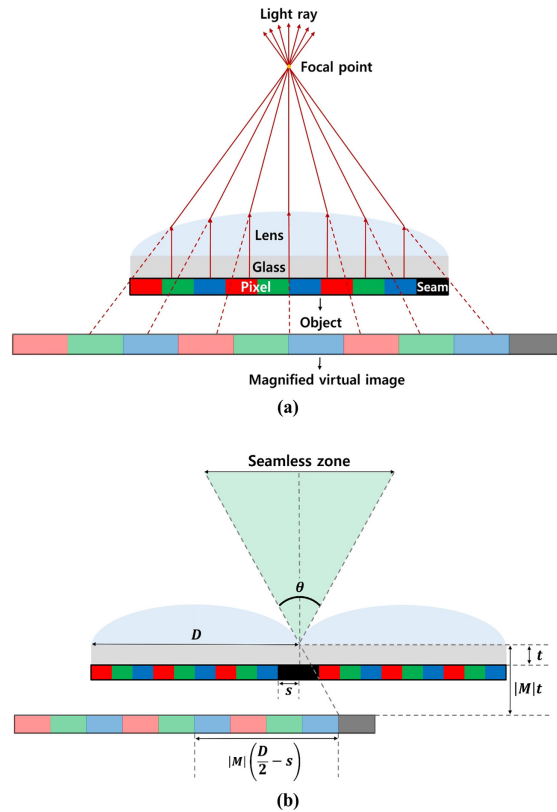


Fig. 4. (a) The image-forming behavior of a plano-convex lens. (b) Principle of seamlessness appearance due to the magnified virtual image of object generated by plano-convex lens pair.

To optimize the physical and optical parameters of CLP for seamless display system, we established the following optimization conditions: (i) Magnification (absolute value) less than 2 (to reduce image distortion), (ii) Viewing angle over 20° (laboratory standard [35]), and (iii) glass thickness from 1 mm to 3 mm (The range of commercially available glass thickness). The focal lengths of each cylindrical lens were calculated from the measured surface profiles by ray optics simulation software (LightTools 8.5, Synopsys). Then, we plotted the calculated viewing angle and the magnification according to the changes of f -number and the thickness of glass under the condition that the seam width is 0.5 mm, as shown in Fig. 5(a) and 5(b). The results showed that the viewing angle increased as the f -number decreased or the thickness of glass increased, as expected from Eq. (4). Based on the optimization conditions and the results shown in Fig. 5, we adopted the lens formed under the dispensing conditions with the moving speed of stage for 20 mm/s under the jetting pressure of 300 kPa without stacked layer and the glass with 2.1 mm thickness for implementing the seamless display. The calculation results for the optimized lens are denoted as red triangle (non-inverted shape) symbols in Fig. 5(a) and 5(b).

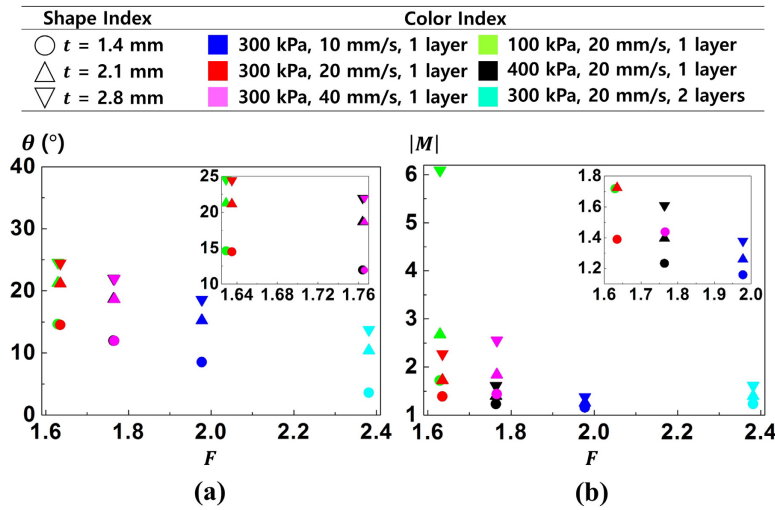


Fig. 5. Calculation results of (a) viewing angle (θ) and (b) magnification ($|M|$) with f -number (F) of the cylindrical lens and the thickness of glass (t). The insets are magnified graphs showing detailed data.

4. Result and discussion

Generally, the luminance is suitable criteria to qualify seamlessness [7,8]. Since the seamless area is relatively brighter than the seam area, there is a luminance difference between the two areas, so that the optical concealment effect by CLP would be investigated by the luminance distribution. For this reason, luminance simulation and measurement of backlight with CLP and those without CLP were performed to investigate the effect of CLP for seamless display. Figures 6(a) and 6(b) are coordinate systems of the backlight without CLP and those with CLP. The center of the seam was set as the origin of the coordinate. First, we conducted the luminance simulation with ray optics simulation software which was utilized for calculating the focal length of the cylindrical lens. The simulated optical system consisted of the light emitting part, seam, glass and CLP with the optimized variables ($F = 1.635$, $|M| = 1.724$, $t = 2.1$ mm, $2s = 0.5$ mm), as shown in Fig. 6(c). As a result, in case of simulation, the variation of luminance was reduced from $\pm 49.76\%$ without CLP to $\pm 1.75\%$ with CLP, as shown in Fig. 6(d). After that, to verify simulation result, we measured the luminance distribution of backlight with CLP and those without CLP. The optical system was constructed by using commercial white light emitting diode (LED) backlight module (MEC-12889, Mechasolution), seam, glass and CLP according to the design variables set on the simulation. The seam was formed by attaching a transparent polymer sheet printed with a black matrix onto the LED backlight and the configured system was mounted to the home-made rotating stage perpendicular to the surface of the optical table for measurement. The luminance was measured along the position of LED backlight in 500 μm intervals in the x -direction by using the luminance meter (CS-200, Konica Minolta) at a distance of 300 mm. At the same x -coordinate, the luminance was measured at three times and the average value was calculated. The uniformity of the measured luminance at the same position was more than 96%. As a result, in case of measurement, the variation of luminance was reduced from $\pm 39.23\%$ without CLP to $\pm 5.44\%$ with CLP, as shown in Fig. 6(d). From both simulation and measurement results, it could be concluded that the luminance was uniformly distributed by adding CLP onto the seam.

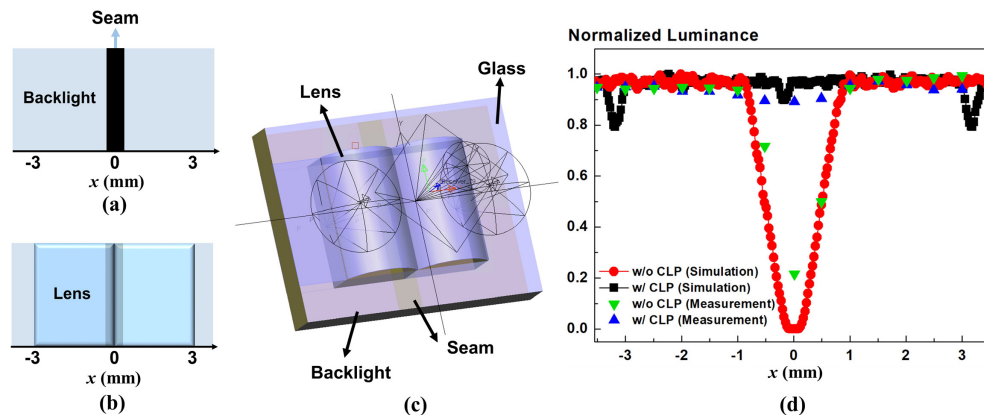


Fig. 6. Coordinate systems of the optical system (a) without CLP and (b) with CLP. (c) The 3-D simulation view of the optical system. (d) Simulation and measurement results of normalized luminance of the optical system with CLP and those without CLP.

According to Fig. 6(d), the luminance of the optical system at the vicinity of CLP boundary ($x = \pm 3$ mm) was reduced by about 20% compared to that inside of CLP in the case of the simulation. This is because a part of the incident light on the top of cylindrical lens surface at the boundary region cannot escape into the air by total reflection [36]. As a result, the luminance is not uniformly distributed and the boundary of CLP appears as a bright strip. To solve this effect, it is possible to consider the method for optimizing the light energy distribution by using freeform lens and diffuser [37].

By assigning the optimized design variables to the Eqs. (1) and (2), which were used in the luminance simulation and measurement, we obtained calculated viewing angle of the seamless display of 21.16° . In order to compare this value with the real case, the viewing angle characteristics of the seamless display was investigated for LED backlight and the actual image. Figure 7 shows seamless display for LED backlight and actual image with different viewing direction. Videos of the seamless LED backlight and actual image are also presented (see [Visualization 1](#) for seamless LED backlight and [Visualization 2](#) for seamless actual image, respectively). In case of using LED backlight, the system configured previously for luminance measurement was used. For actual image, the system was constructed by attaching the glass on which the optimized CLP was formed to the commercial monitor where image with the seam was displayed. The observation distance was 500 mm and the magnification by capturing camera was $4\times$. As expected, the seam was not visible in the viewing range of -10° to $+10^\circ$ (total of 20°) and was still not recognized in the viewing range of -20° to $+20^\circ$ (total of 40°), which is beyond the theoretically expected viewing angle. If the total angular range was wider than 40° , the seam began to be visible, resulting in a wider seam width than actual seam width.

The reason why the seam was not visible at the larger viewing angle than the theoretical value is presumed to be the resolution limitation of the human eye or camera. The angular resolution of a typical human eye is known as about $1/60^\circ$ [38] and that of a camera in case of capturing actual image is about $1/58^\circ$ which can be calculated from the pixel number of image sensor in the horizontal direction (4920), the horizontal field of view (185 mm), the magnification ($4\times$) and the observation distance (500 mm) [39]. If the angular resolution is converted to the distance resolution, it is $145.44\ \mu\text{m}$ for the human eye and $150.40\ \mu\text{m}$ for the camera. Due to this limitation, we concluded that the seam cannot be recognizable to observer even though the observation angle is larger than the theoretical viewing angle until it reaches the angular or distance resolution of observer depending on the observation conditions, according to the previous study on psychophysical requirement for tiled display [40].

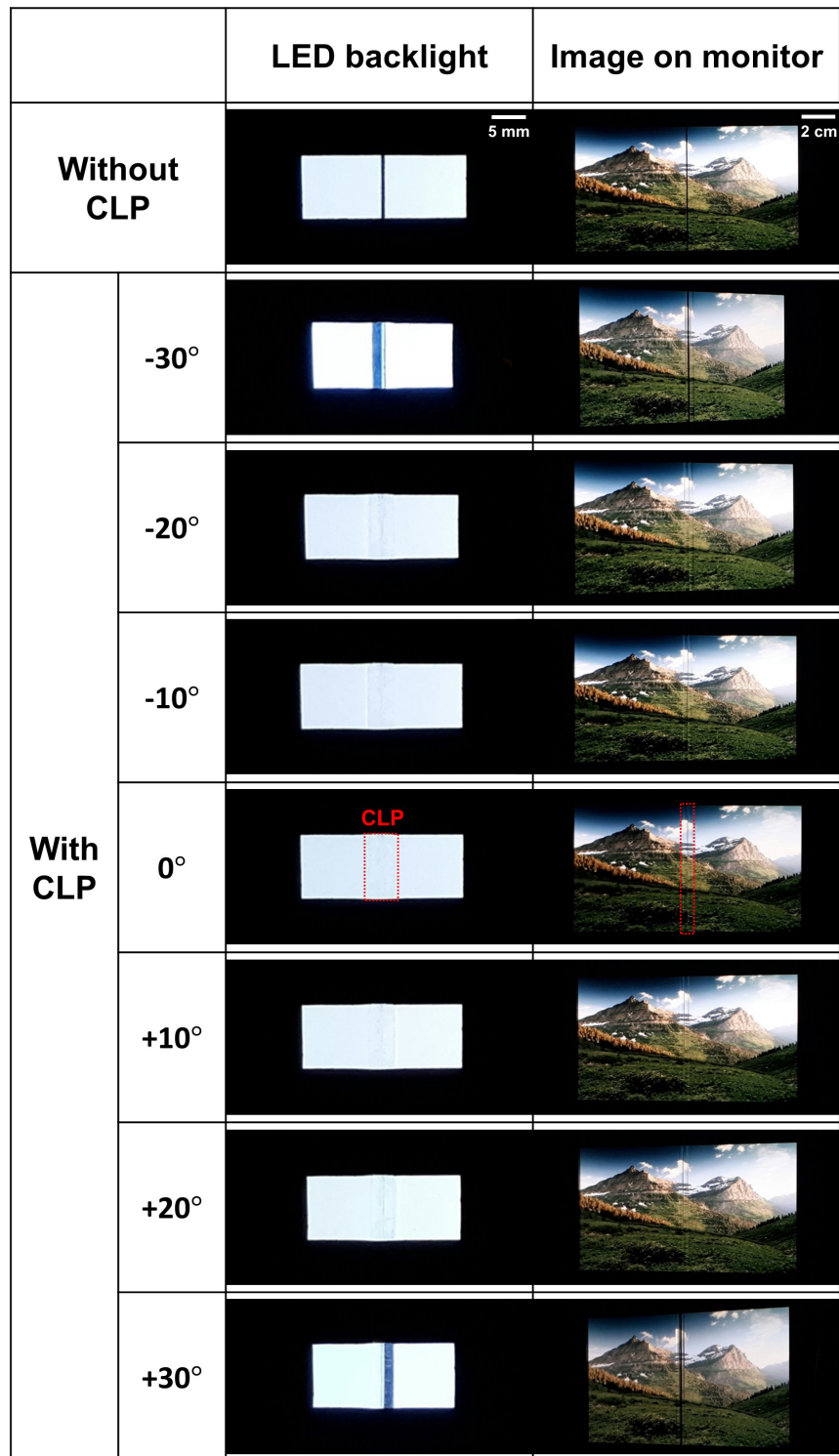


Fig. 7. Seamless LED backlight and actual image on monitor observed from different viewing angle. The positive sign (+) of angles denotes clockwise rotation and the opposite direction is represented as the negative sign (-) of angles when viewed from the front of the image.

Since the fabricated cylindrical lens is a refractive optics element, the original image distortion, which is one of kinds of the Seidel aberration, is caused by the magnification of image [39]. One of the factors that can be optimized to minimize the image distortion at the area of the CLP is image processing algorithm. An approach for the image processing is computational optimization that minimizes the difference between target image (image that desired to be expressed from the display) and actual image (image that actually observed from display), which has been employed for computational light field displays by solving a non-negative least squares problem [41, 42]. Another method is to optimize pixel structure. In this approach, the pixels near the display panel junction are designed smaller in size and higher in luminance than the other pixels [43]. Once the optimization of CLP and proposed approaches are done in the particular display system, it is universally applicable to other identical display system, so the proposed methods can be practical in terms of the development time and mass productivity to solve the image distortion problem.

5. Conclusion

To summarize, we propose a facile method for implementing seamless display simply by forming CLP. The CLP was fabricated by dispensing optically transparent UV curable adhesive with adjusted jetting conditions. Based on the optimized design parameters, we verified the effect of CLP for obtaining seamless characteristic through ray tracing simulation and measurement of luminance distribution. Finally, we demonstrated seamless display with CLP, resulting in a viewing angle of the seam concealment of 40° . We expect that our approach provides a simple and low-cost method for production of seamless tiled public information display. In addition, when the viewing position is fixed close to the display image, such as virtual reality displays and movie theater screen application, our method can easily produce seamless property because the actual viewing angle is relatively small. Furthermore, we believe that the proposed method can be applied not only to flat panel display but also foldable display to remove a seam of the folded part between panels.

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